2.1 Introduction to Optimization

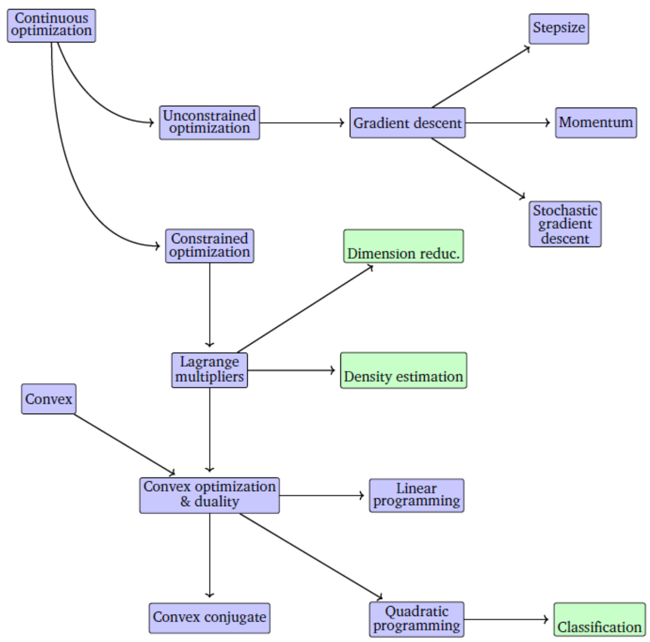
Cost functions, local versus global minima, convexity, and first- and second-order tests.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

The figures follow Deisenroth, Faisal, and Ong, *Mathematics for Machine Learning*.

1. Training is optimization

Training a model is finding a good θ. **Good** is whatever the **cost** (objective) J says it is.



Write $J(\theta): S \to \mathbb{R}$ and solve

$J(\hat{\theta}) = \min_{\theta \in S} J(\theta)$.

Usually $S = \mathbb{R}^l$. This course always **minimizes**. Maximizing profit P is minimizing $-P$.

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2. Linear regression

Model $y = \theta^\top \mathbf{x} + \varepsilon$. Fit by minimizing mean squared error on m training rows:

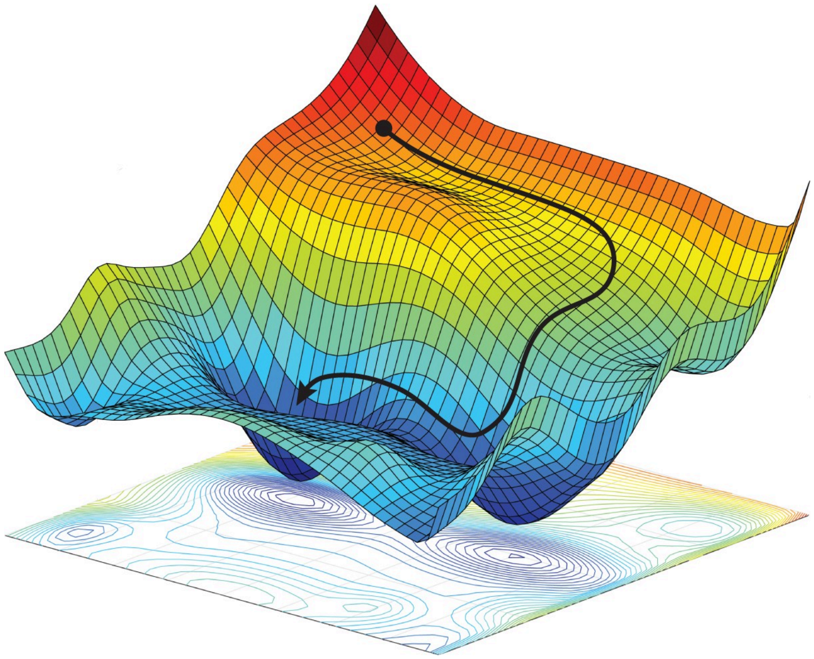
$$J(\theta) = \frac{1}{2m} \sum_{i=1}^{m} \left(\theta^\top x^{(i)} - y^{(i)} \right)^2 .$$

Each step that lowers J shrinks residuals. Better θ on the training set is the hope that predictions on new $\mathbf{x}$ improve too. That hope is generalization; the math this week is the min.

3. Local versus global

A **local** minimum is lowest in a neighborhood: some $\varepsilon > 0$ such that $\|\theta - \hat{\theta}\| < \varepsilon$ implies $J(\hat{\theta}) \leq J(\theta)$. A **global** minimum is lowest on all of S.

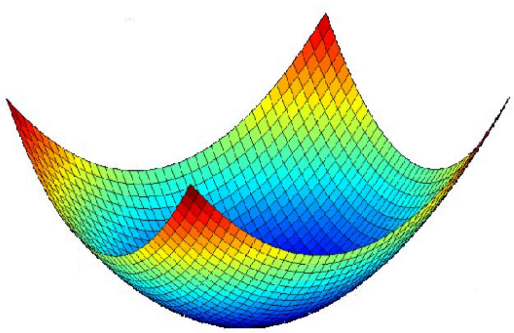
A valley on a hike is local. The lowest point in the whole range is global. Neural-net costs look like the landscape below. Many valleys. The one you land in need not be the lowest.



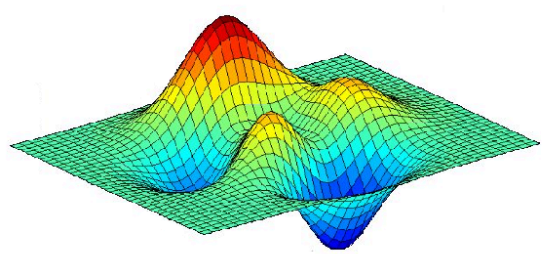
If J is **convex**,

$J\left(t\theta + (1-t)\beta\right) \leq tJ(\theta) + (1-t)J(\beta)$

for all $\theta, \beta \in S$ and all $t \in (0,1)$, then every local min is global.



convex function



non-convex function

Convex J (this course)	Not convex
Linear regression (MSE)	Deep nets
Logistic regression (log loss)	k-means
Soft-margin SVM	GMM likelihood

Convex: gradient descent that reaches a local min has the global min. Nonconvex: you can stop in a valley that is not the bottom.

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4. First- and second-order tests

First-order (necessary). At a local min of a smooth unconstrained J, $\nabla J(\hat{\theta}) = 0$. The slope is flat.

Second-order (sufficient). If that point also has Hessian $\nabla^2 J(\hat{\theta})$ **positive definite**, it is a local min.

A zero gradient is not enough: a saddle is flat and is not a min. Week 2’s algorithms either follow $-\nabla J$ or invert $\nabla^2 J$.

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5. Practice

1. When does a local minimum fail to be global?
2. Why does this course write every learning problem as a **min**, even when the story is “maximize likelihood”?
3. At a point with $\nabla J = 0$, what extra fact about $\nabla^2 J$ lets you call it a local min rather than a saddle?